Unifying the Functions of the Hippocampus

The literature on hippocampal memory functions is already so vast that it now takes many years of study to master. As we have seen, many different theories strive to reduce this vast set of findings into a simpler set of basic principles for hippocampal function. However, there is at this point no consensus on a winner in the war of the theories. How do we make any sense of all these findings?

One perspective that may be helpful is the evolutionary one. The brain is fundamentally a predictive organ, designed to use sensory input and past experience to help the organism make good guesses about what future behavior will give the best chance of survival and eventual reproduction. Outside the laboratory setting, the means of meeting our needs (food, water, shelter, mates) are rarely close at hand. Instead, many organisms have a territory, and this territory is filled with potential resources that may be useful when the organism becomes hungry, thirsty, cold, or pursued. Getting to these resources requires a territory map, and this map must be constantly updated with new events to stay accurate. So spatial maps require episodic memories to be useful for survival. The unique circuitry of the hippocampus allows for the rapid learning and high information capacity needed to make the whole system work effectively (Amaral, 1993; Lynch, 2004).

One implication of this view is that what we call an episodic "memory" system actually serves not only to record the events of the past, but also to make predictions about possible events in the future. A thirsty animal in the desert must recall past watering holes, but also must predict which of these would be most likely to provide water in the near future. Interestingly, this is exactly the finding of recent studies: the hippocampus and related systems are as much involved in imagining the future as in remembering the past (Squire et al., 2010). In the next section, we will take a closer look at the role of these areas in future memory.

Remembering the Future: Prospection and Imagination

Are you hungry right now? What would you most like to eat? Perhaps a plate of hot, salty fries? Or a grilled sandwich with fresh sliced tomato, basil, and mozzarella? Do you plan to make this tasty meal on your own or go out and buy it? Let's say you decide to buy it. Where are you going to go? On the one hand, you'll need a map of all the nearby restaurants, snack bars, cafeterias, and other sources of ready-made food. Then, you'll need to decide which one would most likely satisfy your current craving. To plan your excursion, you'll need to go through your past experiences of meals at the places on your map. One snack bar is nearby and inexpensive, but the food there was greasy and left you feeling ill in the past. Another makes your favorite sandwich, but always has a long line. Thinking ahead, you're not sure you want to wait that long. In fact, right now, you feel so hungry that the thought of eating a big plate of greasy fries seems quite appealing. Putting together the experiences of your recalled past and your imagined future, you decide on the cheap snack bar, consult your spatial map for directions, and head out the door.

If you play through this little scenario in your mind's eye, you will see immediately how our remembered past and imagined future are closely interwoven in helping us decide what to do to meet our needs. However, only recently have we discovered that the brain's episodic memory systems are just as important for imagining future experiences as they are for imagining past experiences. The term **prospection** has been coined to describe the process of "remembering the future" (Ingvar, 1985), just as the term **recollection** describes remembering the past. The influential memory researcher Endel Tulving once described the function of episodic memory as "mental time travel" (Tulving, 1985). In this section, we will see how episodic memory systems engage in time travel to the future as well as the past.

How We Imagine Future Experiences

In 2007, Demis Hassabis and other researchers at the Institute of Neurology in London made the striking observation that patients with hippocampal lesions not only had amnesia for past experiences, but also could not imagine new experiences (Hassabis, Kumaran, Vann, & Maguire, 2007). For example, when asked to imagine standing in a museum full of exhibits, one patient stated "there's not a lot coming . . . it's not very real . . . I'm not picturing anything . . . I'm not imagining it." In other scenarios, such as lying on a beach, patients might describe a blue sky or a few isolated sounds. Even providing a patient with relevant source material such as pictures, sounds, and smells did not help them imagine the scenes.

Unlike that of healthy controls, the patients' imagery lacked the rich, vivid details of a realistic episode. The patients' descriptions were less detailed than those of control subjects in many categories: spatial references, number of entities present, sensory descriptions, and thoughts, emotions, and actions. The fundamental problem seemed to be a lack of spatial coherence: the imagined experiences were a collection of fragmentary sensations rather than a unified episode in a particular setting (Hassabis et al., 2007).

Neuroimaging studies have provided strong evidence that the brain uses a core network of regions both for remembering past experiences and for imagining new or hypothetical experiences (FIGURE 9.11) (Schacter, Addis, & Buckner, 2008). Yet the hippocampus and parahippocampal cortex are merely one part of this network. Also involved are areas well outside

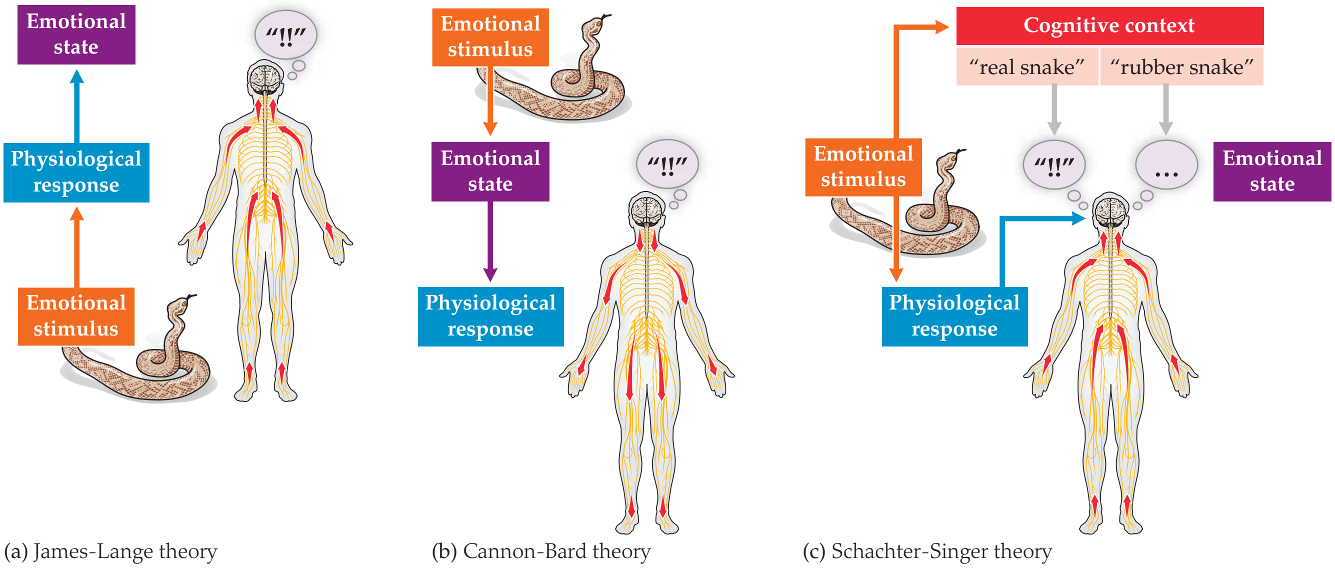

FIGURE 13.9 Summary of the three theories of emotion. (a) The James–Lange theory, a bottom-up theory; (b) the Cannon–Bard theory, a top-down theory; and (c) the Schachter–Singer theory, a two-factor theory.

interviewed people who crossed a much less frightening, lower, sturdier bridge nearby. For the study, subjects were not recruited to participate, but, rather, researchers interviewed people who had just crossed the bridge as part of their normal daily life. A research assistant simply stood at one end of the bridge and asked young men who had just crossed the bridge whether they would mind filling out a survey, in which they had to write a brief story. The assistants then tore off a corner of the survey and wrote down a fictitious personal name and an actual telephone number at which they could be reached, handed the piece of paper back to the participants, and invited the participants to call if they wanted to talk further.

After crossing the lower, sturdier bridge, only a small minority of the (all male) participants actually called the number of the female assistant. However, after crossing the high suspension bridge, the majority of the subjects called the female assistant. Their stories also showed stronger ratings for sexual imagery than the stories of those who had crossed the lower bridge. Neither of these effects turned up when the assistant had been male. The male assistant received calls from fewer than 10% of subjects who crossed either type of bridge. These findings were taken as further evidence for both the two-factor model of emotion and the misattribution of physiological arousal to emotional state: in this case, sexual attraction rather than fear. (See **FIGURE 13.9** for a comparison of the three theories of emotion discussed so far.)

How far have our theories of emotion come since the two-factor models of the 1970s? Over the past four decades, three major trends have taken place among theories of emotion. First, most of the modern accounts of emotion incorporate both bottom-up visceral factors and top-down contextual factors, although the degree of emphasis varies between these two poles. Second, many modern accounts have moved away from single-system explanations of emotion and toward multiple, partly dissociable systems to supply the rich array of colors in our emotional palette, often drawing on studies of aberrant emotion in people with neurological illnesses, as we'll see in some examples later in this chapter. Third, the accounts themselves have become more and more anatomically specific in their descriptions of the underlying neural circuitry of emotions.

Core Limbic Structures: Amygdala and Hypothalamus

To connect the emotional responses to the appropriate inner states, we must add another layer to the neural hierarchy of emotion. This layer will draw on a much more sophisticated set of inner-world and outer-world sensory inputs than the brainstem: visceral sensations, hormone levels, and serum concentrations of key electrolytes and biochemical compounds, as well as visual input, auditory input, somatosensory input, olfactory input, and input from all the other external-world senses. Thanks to these rich and diverse inputs, the neurons in this next layer will be in a much better position to determine the organism's basic survival needs and to promote the appropriate behaviors to address them. This layer encompasses the core structures of the limbic system, and two of its most important centers are the hypothalamus and the amygdala.

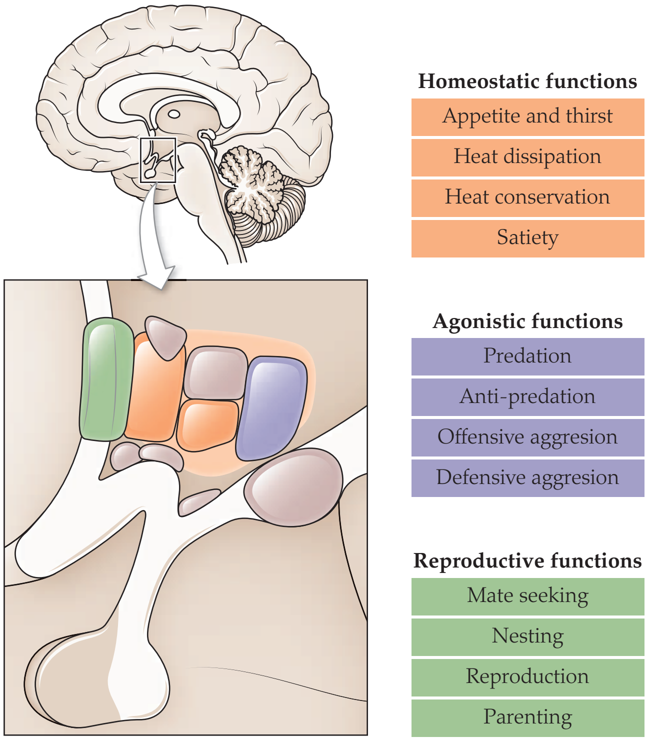

FIGURE 13.10 The hypothalamus and survival drives. The nuclei of the hypothalamus coordinate core survival functions, including homeostatic drives, agonistic drives, and reproductive drives.

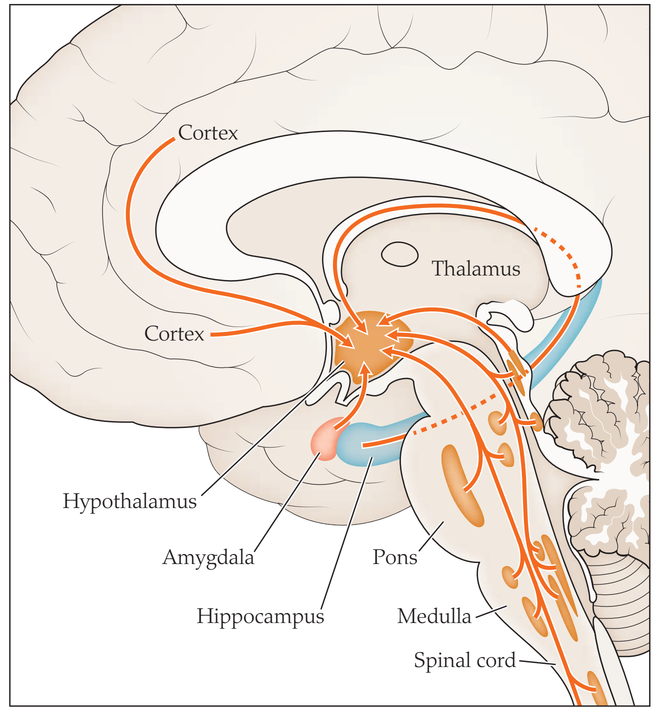

FIGURE 13.11 Major input pathways to the hypothalamus. Inputs come from the visceral sensory neurons of the spinal cord, visceral sensory nuclei within the brainstem, limbic regions of the cortex, and from the hippocampus and amygdala.

Hypothalamus: Internal States, Homeostatic Drives

The hypothalamus lies just below the thalamus (FIGURE 13.10) and is highly conserved in structure across a wide variety of animal species, from hagfish to human beings (Sower, Freamat, & Kavanaugh, 2009). This commonality of structure stems from a commonality of purpose: all living animals must take care of reproductive, appetitive, and agonistic functions to survive. Reproductive behaviors are essential to every animal that wants to keep its genes in circulation. Appetitive behaviors, like feeding and drinking, are essential to every animal that wants to survive long enough to reproduce. Finally, so-called agonistic behaviors (attack and defense) are essential to every animal that wants to avoid being eaten or killed by its competitors. Whether you swim, fly, or ride the bus, your brain must be able to perform these three classes of motivated behavior.

Appetitive behaviors help the body meet its basic needs for energy, water, electrolytes, temperature maintenance, and so on. The process of keeping these critical internal-world parameters in balance is called homeostasis, and the maintenance of homeostasis is one of the key functions of the hypothalamus. To keep track of the body's internal state, the hypothalamus draws on the same visceral sensory input pathways that we have been tracing up from the internal organs through the spinal cord and brainstem (FIGURE 13.11).

Yet the hypothalamus also has a second major source of input that is unavailable to most of the brainstem nuclei: sensors that monitor the composition of the bloodstream itself. This is a highly unusual feature. The vast majority of the brain's neurons hide behind a sort of cellular insulation known as the **blood–brain barrier**, which keeps them in an optimized chemical environment free from noxious substances. Yet there are a handful of regions where the blood–brain barrier is porous, allowing neurons to peek out and monitor the composition of the bloodstream. Some can sense the bloodstream's concentration of electrolytes, to monitor water balance. Others can sense the levels of glucose, insulin, or other markers of energy balance. Still others can identify circulating levels of various hormones: stress hormones like cortisol, female sex hormones (**estrogens**) like **estradiol**, male sex hormones (**androgens**) like **testosterone**, metabolic hormones like **thyroid-stimulating hormone** and **growth hormone**, and so on. Others even monitor the blood for markers of infection and immune system activity like **cytokines**, interleukins, or complement proteins. So the hypothalamus can sense when you are overfed, underfed, dehydrated, lacking in glucose, coming down with the flu, or ovulating—to name just a few of its interoceptive abilities (FIGURE 13.12).

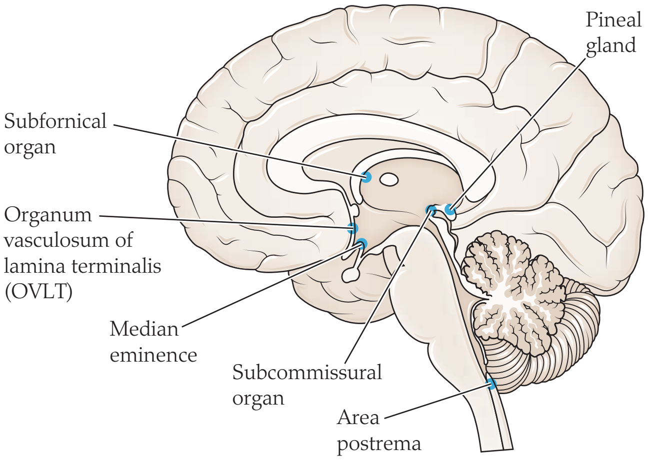

FIGURE 13.12 Regions of the brain where the blood–brain barrier is more permeable. These regions act as "windows" through which the hypothalamus and other brain regions can sense the chemical and hormonal composition of the blood, to monitor the internal environment.

These kinds of internal sensory inputs are critical for guiding the appetitive categories of behavior: foraging, feeding, drinking, staying warm, and so on. Some of the dozens of nuclei in the hypothalamus serve appetitive functions. For example, the **lateral hypothalamus** regulates hunger and is counterbalanced by a **ventromedial hypothalamus** that regulates satiety. An anterior nucleus of the hypothalamus regulates heat dissipation, whereas a posterior nucleus regulates heat conservation. Activity in a tuberomammillary nucleus promotes wakefulness, whereas activity in a ventrolateral preoptic area promotes sleep.

The hypothalamus also plays a key role in regulating reproductive functions, including sexual behavior and parenting behavior. The size and structure of the ventromedial hypothalamus are different in males and females. Neurons in this region direct a variety of sexual behaviors, such as copulating, vocalizing, and scent marking (Harding & McGinnis, 2004). They also drive some sex-specific behaviors, such as the female-specific sexual posture of **lordosis** in rats (Pfaff & Sakuma, 1979). Another region, called the **medial preoptic area**, is involved in male copulatory behaviors: erection, intercourse, and ejaculation. In females, however, this same nucleus is several times smaller and instead regulates maternal-specific behaviors: keeping the infant close by, nursing the infant, grooming and cleaning the infant, and so forth.

Agonistic functions also fall under the purview of the hypothalamus. Both aggressive and defensive behaviors can be elicited by stimulation of various hypothalamic nuclei, including the anterior hypothalamic nucleus (Adams, 2006; Adams et al., 1993), the dorsal premammillary nucleus (Canteras, Chiavegatto, Ribeiro do Valle, & Swanson, 1997), and the ventromedial nucleus (Lin et al., 2011; Martinez, Carvalho-Netto, Amaral, Nunes-de-Souza, & Canteras,

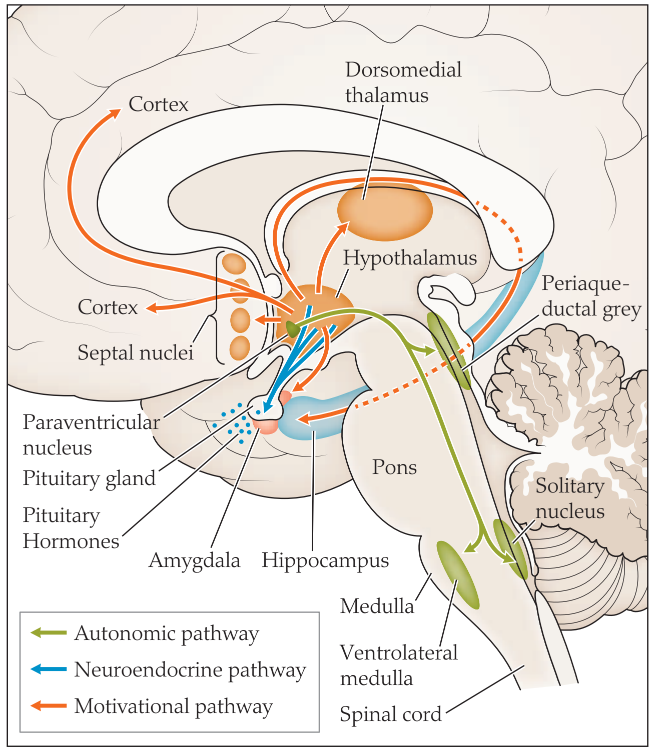

FIGURE 13.13 Three output pathways from the hypothalamus. An autonomic pathway regulates the sympathetic and parasympathetic nervous system; a neuroendocrine pathway regulates hormone secretion in the pituitary gland; a motivational pathway up through to the cerebral cortex sets behavioral priorities.

2008). We will explore the role of the ventromedial nucleus further in a case study later in the chapter.

To effect changes in one's internal state, the hypothalamus has three output pathways available (FIGURE 13.13). First, an autonomic output pathway via the **paraventricular nucleus** serves to stimulate the sympathetic and parasympathetic nervous system. For example, low glucose can prioritize feeding activities like salivation, biting, chewing, and swallowing (Cai et al., 1999; Tolle et al., 2002). So the hypothalamus has pathways capable of deploying simple brainstem action plans when internal conditions require them.

Another output pathway, the neuroendocrine pathway, regulates hormone levels. The hypothalamus is often called the master control gland or master endocrine gland because it controls the activity of every other **endocrine gland** in the body. For example, in response to various kinds of stress, the hypothalamus secretes **corticotropin-releasing hormone**, which passes into the pituitary, causing it to release **adrenocorticotropic hormone (ACTH)**, which in turn passes to the body's adrenal glands to stimulate the production of the so-called stress hormone, cortisol. Cortisol itself has a wide variety of bodily effects: suppressing the immune system, increasing blood sugar, and altering metabolism to promote fat storage and weight gain. Many varieties of stress can

CASE STUDY:

An Internal Growth of Rage

In 1962, a 20-year-old bookkeeper in New York was admitted to the hospital with some unusual changes in her behavior (Reeves & Plum, 1969). For the past year, she had begun eating ever-increasing amounts of food and drinking increasing amounts of fluid, far beyond her usual intake. Her menses had stopped altogether, and she complained of frequent headaches.

Her doctors suspected that she had a brain tumor, but in those days, before the invention of CT or MRI scans, brain imaging techniques were relatively crude. To outline the structures of the brain, they performed a pneumoencephalogram. This involved literally filling the brain's ventricles with air rather than cerebrospinal fluid and then taking an X-ray to examine the ventricles' appearance, as the doctors would with a pair of air-filled lungs. On the X-ray, they saw evidence of a growth near the hypothalamus and performed a surgical "exploration" of the patient's brain in an attempt to find and remove the suspected tumor. Unfortunately, the surgeons saw no obvious abnormality, and the patient soon recovered and went home untreated.

However, nearly two years later, the patient returned, this time with symptoms much more severe (Reeves & Plum, 1969). She was socially withdrawn, but would explode into fits of laughing, crying, and rage without obvious reasons. However, unlike in pseudobulbar affect, this was more than a mere reflex reaction. In her irritable state, the patient would refuse to cooperate with simple requests or would scratch and even attempt to bite her examining physician. She was admitted to the hospital, where she persisted in hitting, biting, scratching, or throwing objects at people around her. This behavior was not entirely automatic, but internally driven: she had taken to eating 8,000–10,000 food calories a day and would invariably attack her attendants when they refused to give her any further food. Only when given food nearly continuously would she desist from her aggressive behavior.

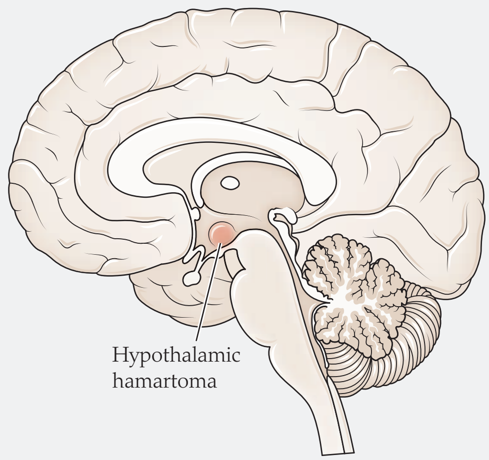

FIGURE 13.14 **Hypothalamic hamartoma.** This very rare tumor can cause fits of laughter, crying, or aggressive behavior by triggering abnormal bursts of activity in the nuclei of the hypothalamus.

Once again, surgical exploration was performed. This time, the surgeons found a tumor in the expected area, near the ventromedial hypothalamus. However, because of its location, they were unable to remove it without endangering her life. With the growth in her hypothalamus, other key functions became impaired: her blood sugar rose into the diabetic range, her thyroid hormone levels dropped, she began spiking fevers without external cause, and her electrolytes fell rapidly out of balance. Three weeks after the operation, she developed a critically high level of sodium in her blood and died.

A postmortem examination found a rare kind of tumor in the ventromedial hypothalamus. The effects of a hypothalamic tumor are quite unusual: changes in eating behavior, precocious puberty and other hormonal changes, fits of laughter, and, often, fits of inappropriate rage (FIGURE 13.14) (Veendrick-Meekes, Verhoeven, van Erp, van Blarikom, & Tuinier, 2007). Some patients develop symptoms of depressed mood, or anxiety, or rapid mood swings (Veendrick-Meekes et al., 2007). As with the woman in this case, the disturbances typically have reportable, internal emotional content. Unlike in pseudobulbar affect, the manner of emotional expression can be quite variable (Lieberman & Benson, 1977; Veendrick-Meekes et al., 2007). Rage may be expressed verbally, physically, or through passive resistance and withdrawal. The symptoms of these patients are not empty automatism, but real subjective emotional states, driven by damage to a specific pathway in the circuitry of the brain.

produce increased cortisol: illness, fever, injury, pain, and physical exertion, as well as the emotions of anxiety or depression (Maglione-Garves, Kravitz, & Schneider, 2005). Conversely, cortisol itself can affect mood, often producing a sense of euphoria when administered as a medication (Ling et al., 1981). So the hypothalamus provides a key pathway that links hormonal activity to emotional state. Similar links between hormone and mood can be seen for thyroid-stimulating hormone and some sex hormones, including the estrogens and androgens.

The third pathway is motivational. If glucose levels are dropping, all of the autonomic and hormonal adjustments in the world are no substitute for finding and ingesting a good square meal. Getting a meal, or a drink, or a mate is no simple feat. A whole series of problems must be solved: evaluation, goal-setting, strategy formation, behavior selection, action planning, and the matching of actions to the surrounding environment. Each one of these steps requires far more complex circuitry than the hypothalamus can muster on its own. Fortunately, the hypothalamus has a third output pathway that leads to forebrain structures like the striatum and cerebral cortex: areas capable of putting together complex, flexible, nuanced behaviors.

Do Hypothalamic Circuits Generate Inner Emotional Experiences?

Up until now in the hierarchy, we have seen circuits that generate the building blocks of outward emotional expression, but not the actual inner experience of emotion. The brainstem can produce sham rage or mirthless laughter, but not the real thing. Can the hypothalamus do any better? Does activity here also generate a subjectively empty behavior or a complete emotion with both subjective experience and outward manifestations?

Case reports of patients undergoing therapeutic deep brain stimulation (DBS) can offer us some clues. In one case, inadvertent stimulation of the posterior medial hypothalamus caused the patient to develop an instantaneous attack of emotional rage, agitation, and shouting. In the middle of the surgery, the patient attempted to forcibly remove the stereotactic frame from his own skull, which might have been a lethal act if he had succeeded. Fortunately, he was physically restrained and briefly sedated. After the sedation cleared, the patient recalled that he had felt angry and acted aggressively, but could not explain why (Bejjani et al., 2002).

In another case, a 50-year-old woman underwent DBS implantation for obesity. Here, stimulation of the ventromedial hypothalamus produced all the features of a panic attack, complete with physiological effects (hyperventilation, increased blood pressure, and increased heart rate), interoceptive features (nausea and a sensation of shortness of breath), and subjective emotional experiences of overwhelming anxiety (Wilent et al., 2010).

Why should subjective emotional experiences emerge at this level of the hierarchy and not in others? Although no one is sure so far, we can make note that the hypothalamus is one of the first levels of the hierarchy that ties together three functions: the perception of inputs relevant to survival drives, the representation of internal drive states engendered by these inputs, and a process for modifying those internal states through changes in physiology and motivation. These three processes seem to be common to most of the core circuitry of subjective emotion throughout the brain.

Amygdala: Externally Generated States and Drives

The amygdala is another core limbic structure, like the nearby hypothalamus. But where the hypothalamus looks inward to the body, the amygdala looks outward to the world. Its sensory sources come from visual inputs, auditory inputs, and olfactory inputs via the thalamus and cortex. Some interoceptive information is also available via the ascending pathways through the brainstem (**FIGURE 13.15a**). The amygdala's output pathways lead in three main directions: down to the somatic and autonomic nuclei in the brainstem, over to the hypothalamic nuclei responsible for hormone secretion and for the three main classes of motivated behaviors, and up via the striatum and other ascending pathways to large swaths of the cortex involved in evaluation, motivation, and behavioral control.

Like the hypothalamus, the amygdala is organized into a collection of more than a dozen different nuclei (**FIGURE 13.15b**), all richly interconnected, most of whose functions are still being worked out. Some of these nuclei fall into a subdivision known as the **basolateral amygdala**, whereas others fall into a subdivision known as the central amygdala (Cardinal, Parkinson, Hall, & Everitt, 2002).

The basolateral amygdala has its strongest connections with higher sensory regions in the cortex, and its microscopic structure is also more akin to that seen in the cortex. Its role is sometimes summarized as a tracker of value because its neurons are capable of learning and tracking the emotional significance of stimuli from the outside world, drawing on every sensory modality: visual, auditory, somatosensory, gustatory, and olfactory (Pickens et al., 2003; Zald, 2003).

The **centromedial amygdala**, in contrast, has its strongest connections with regions further down the hypothalamus and brainstem. Its role is sometimes summarized as a controller of these regions (Phillips & LeDoux, 1992). As the output rather than the input side of the amygdala, it draws on external-world sensory information to modulate the internal drives and hormonal functions of the hypothalamus (Pessoa, 2010).

These two regions are closely interlinked and serve complementary functions. Where the basolateral amygdala is

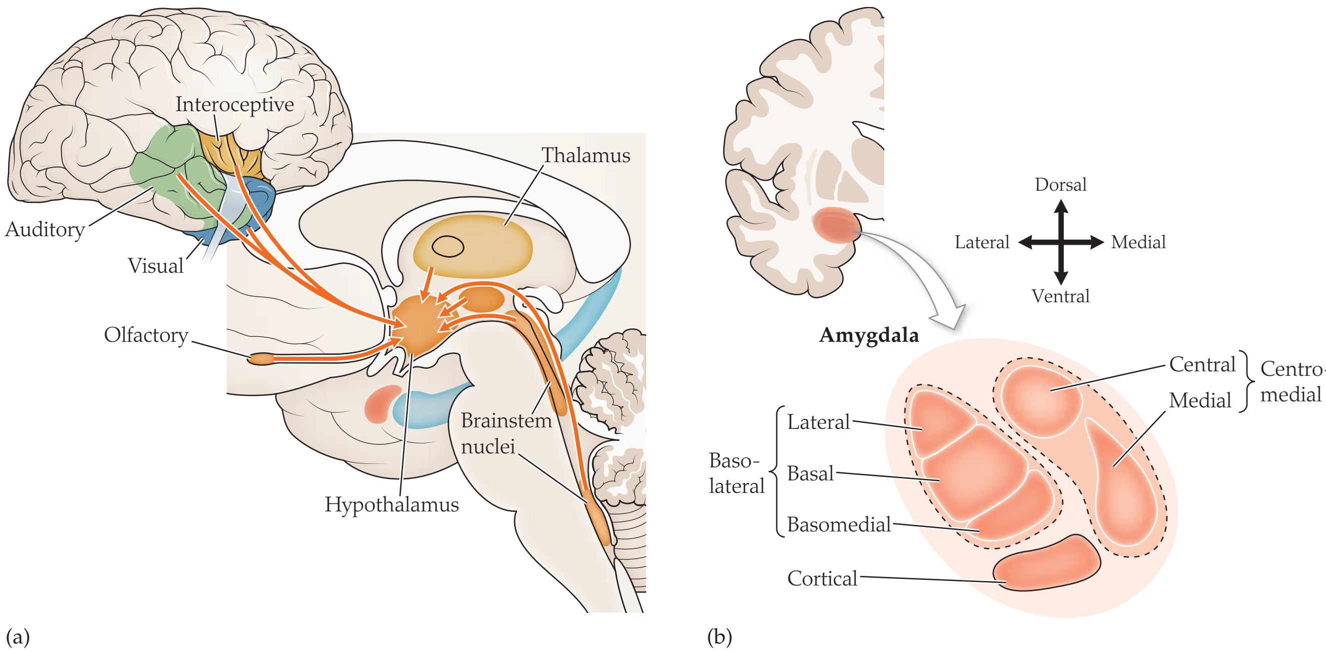

FIGURE 13.15 The amygdala. (a) Input pathways to the amygdala. (b) Some of the major subdivisions of the amygdala, including the basolateral nuclei and the centromedial and cortical nuclei.

wired to interpreting the emotional value of a stimulus, the centromedial amygdala is wired to implementing an appropriate emotional response, with autonomic, hormonal, and behavioral components. To paraphrase one author, the amygdala proceeds from the question of "what is it worth?" to the question of "what is to be done?" (Pessoa, 2010).

One of the key emotional roles for the amygdala lies in the detection of threats and the implementation of fear responses (LeDoux, 2007). For example, in the classical **fear conditioning** paradigms of behaviorists, animals are exposed to an auditory tone, which is soon followed by a painful electrical shock. Over time, the animals learn the association between the unconditioned stimulus of the shock and the conditioned stimulus of the auditory tone. Eventually, they begin to exhibit fear responses to the tone itself: freezing, increases in heart rate, and so on.

The amygdala is critical for conditioned fear learning. Through a circuit that is now quite well studied, the unconditioned stimulus of the shock activates pain receptors in the skin, which send signals via the thalamus and somatosensory cortex to the **lateral nucleus** of the amygdala (Phillips & LeDoux, 1992). Signals then pass to the centromedial amygdala to coordinate the appropriate fear response: freezing, stress hormone secretion, increased heart rate, and so forth. As for the conditioned stimulus of the auditory tone, this produces activity in auditory regions of the thalamus and cortex, which stimulates neurons within the lateral nucleus of the amygdala (Phillips & LeDoux, 1992). Through the learning mechanisms of synaptic plasticity and long-term potentiation (Chapter 9), the connections between these neurons and the output neurons in the central nucleus are gradually strengthened. In time, the auditory stimulus alone becomes capable of driving the fear responses of the central nucleus, through the newly strengthened pathways (FIGURE 13.16) (Rogan & LeDoux, 1995).

Some other amygdala neurons track positive value, or reward, rather than negative value, or aversiveness. In another type of conditioning experiment, animals learn to associate a conditioned stimulus (such as a colored box) with a rewarding unconditioned stimulus (such as a tasty piece of food inside the box). Just as some networks of neurons in the amygdala can learn and unlearn the "threat value" of a stimulus, so other networks can learn and unlearn the "reward value" of a stimulus. They can even adjust the reward value based on the organism's internal state. For example, if we offer an animal a choice between two different boxes with two different foods, A and B, the neuron may respond equally to either box. But if we first feed the animal to satiety on one food, the neuron may cease to respond to the box containing that food, while continuing to respond to the box with the other food inside. So amygdala neurons can take into account the organism's needs when calculating the value of a stimulus in the outside environment (Pessoa & Engelmann, 2010).

The Amygdala and Emotional Experience

Like hypothalamic lesions, amygdala lesions have profound impacts on behavior and emotion. In the 1930s, the psychologist Heinrich Kluver and neurosurgeon Paul Bucy studied the effects of removing the anterior temporal lobes (including the amygdala) in rhesus monkeys (Kluver &

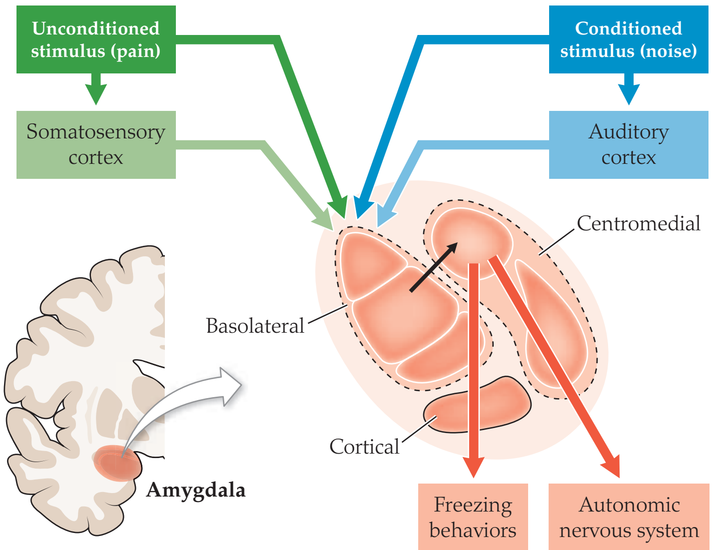

FIGURE 13.16 The classical conditioning pathways in the amygdala. An unconditioned stimulus (US), pain, and a conditioned stimulus (CS), a noise, are presented together. Both the US and the CS stimulate the basolateral amygdala and are paired together, stimulating the central nucleus of the amygdala. The output from the central nucleus triggers the freezing behaviors and activation of the autonomic nervous system. Over a few trials of learning, the connections between these nuclei are strengthened, so that the noise comes to trigger the response on its own.

CASE STUDY:
The Woman Who Knows No Fear

Of the more than 7 billion people in the world, fewer than 300 are known to have the rare illness known as **Urbach–Wiethe disease** (Feinstein, Adolphs, Damasio, & Tranel, 2011). In addition to developing lesions throughout their skin and mucous membranes, these patients sometimes suffer destructive lesions through their medial temporal lobes: a region containing the hippocampus and amygdala. In even rarer cases, the amygdala alone is destroyed, whereas the rest of the brain remains intact (Sainani, Muralidhar, Parthiban, & Vijayalakshmi, 2011).

The most famous of all such cases is a woman known by the initials S. M., the so-called "woman without fear" (Young, 2010). She lacks an amygdala on either side, but her brain MRI shows no other lesions (FIGURE 13.17). Although her general intelligence, memory, language, and perceptual functions all fall within the normal range, she has one major difficulty: a severe impairment in the ability to either experience or recognize the emotion of fear. When viewing emotional film clips, she experiences all the rest of the usual palette of human emotions: happiness, sadness, disgust, anger, and surprise, all expressed in the usual human ways. With frightening films, however, her experience is one not of fear, but of excitement and exhilaration.

A recent study of her real-life experiences makes for astonishing reading (Young, 2010). Despite an avowed fear of snakes, when brought to a pet store she rushed over to the snakes, overcome by a feeling that she described as excitement and curiosity rather than fear. She held nonpoisonous snakes, stroked them, and touched their tongues. She reported feeling compelled to poke and touch even the large, dangerous snakes in the store and asked over and over again for permission to do so.

The next visit was to a famous "haunted house," the Waverly Hills Sanatorium. S. M. spontaneously led a group of visitors through its darkened hallways, filled with hidden actors in monster costumes. As the actors sprang out from their hiding places, the other group members screamed loudly. In contrast, S. M. smiled or laughed, approached the monsters, or tried to start conversations with them. At one point she even frightened one of the monsters by giving the actor a firm poke in the head "to see what it would feel like."

S. M.'s altered emotional repertoire has had a significant impact on her life. Her inability to detect or respond to environmental threats or to learn from past experience has placed her in grave danger many times. Living in a poor urban area, she has been held at knifepoint and gunpoint, physically attacked by a much larger woman, explicitly threatened with death,

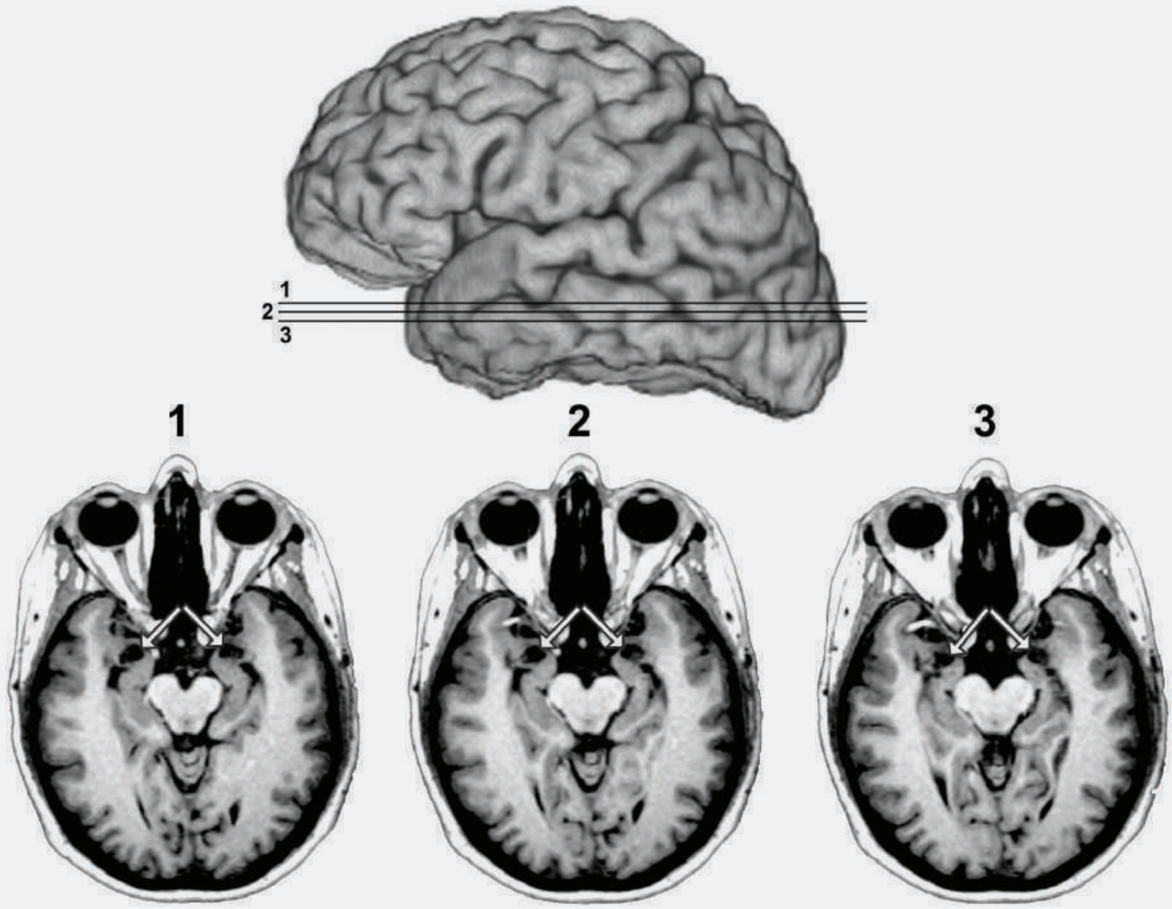

FIGURE 13.17 **MRI scans of Patient S.M.'s brain.** The arrows point to the amygdala, the region of the brain that S.M. is missing, as shown by the vacant black holes underneath the arrows.

and nearly killed in an act of domestic violence. In one case, after being physically attacked in a park, she returned to the same park the very next day!

S. M.'s rare case sheds new light on the contentiously debated role of the amygdala in human emotion. The loss of the amygdala has not removed her capacity for all emotions, but has had specific effects on her ability to evaluate threats and to deploy the sense of fear as appropriate. In the absence of fear, excitement and curiosity combine to produce approach rather than retreat, with consequences that have proved nearly lethal on more than one occasion. As the authors conclude in their report, "it appears that without the amygdala, the evolutionary value of fear is lost" (Young, 2010).

Bucy, 1939/1997). The monkeys developed a sort of fearless curiosity in exploring their surroundings. This was accompanied by **visual agnosia** (an inability to interpret what they saw before them), which has been explained by damage to the ventral visual stream (Chapter 5). Also, the monkeys displayed inappropriate forms of motivated behavior: they showed hyperorality (an indiscriminate tendency to place objects in their mouths, whether edible or not). In addition, they showed hypersexuality, masturbating constantly and attempting to copulate indiscriminately, even with objects or members of other species. This constellation of symptoms was termed **Kluver–Bucy syndrome**.

In human beings, similar symptoms were later reported in cases of widespread lesions of the anterior temporal lobes and amygdala (Terzian & Ore, 1955). However, the most recent studies of humans with isolated amygdala lesions suggest that the full Kluver–Bucy syndrome does not arise in humans (Hayman, Rexer, Pavol, Strite, & Meyers, 1998). Instead, such patients have mostly normal emotional functioning, with the exception of severe impairments in the learning and expression of fear.

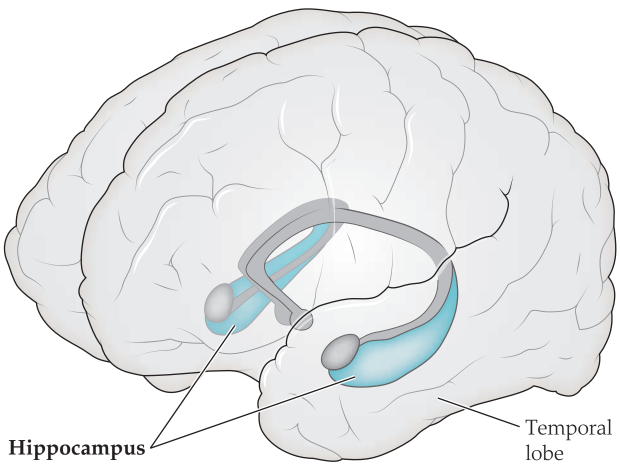

FIGURE 13.18 **The location of the hippocampus within the temporal lobe.**

Hippocampus: Emotional Memories

As we saw in Chapter 9, the hippocampus is a key brain region for encoding and retrieving certain kinds of memories. As you may recall, the hippocampus (FIGURE 13.18) seems to perform at least two different kinds of functions: spatial navigation and episodic memory. Although on the surface these two functions may seem quite different, in fact they are tightly interrelated. For example, it can be delightful to recall spending an afternoon in a beautiful, atmospheric café in a tiny square in a foreign city, but it can also be frustrating if you return to that city a year later and cannot find your way there again. Conversely, it can be helpful to remember the location of a secret shortcut to your place of work, but if a new construction project blocks your shortcut, you must update your map of the territory using episodic memory or else risk a long detour. If you have ever used a Web-based map service to find a store or restaurant, only to arrive there and find that it closed six months ago, you can appreciate the need for episodic memories to keep your maps up to date.

So, emotional, episodic memories and spatial navigation are interrelated functions. How does a single brain region look after both of these functions at once? As it turns out, the anatomical structure we call the hippocampus may actually consist of at least two (and possibly more) distinct subregions (Moser & Moser, 1998). Although invisible to the naked eye, these subregions are distinguished by completely different patterns of connectivity to the cortex and elsewhere in the brain. They also have different patterns of gene expression within their neurons (Fanselow & Dong, 2010). Furthermore, they seem to have distinct functional roles, as revealed by lesion and neuroimaging studies.

In humans, the posterior hippocampus seems to specialize in the spatial component of memories. In neuroimaging studies, the posterior hippocampus activates during tasks of spatial navigation and spatial memory recall (Maguire, Valentine, Wilding, & Kapur, 2003). Compared with novice taxi drivers, experienced London taxi drivers, who as part of their profession must store a vast amount of spatial and navigational knowledge, show increases in hippocampal size that are specific to the posterior rather than the anterior hippocampus (Maguire et al., 2000).

In contrast, the anterior hippocampus has a much stronger role in emotional memory than in navigation. Where lesions of the posterior hippocampus interfere with spatial navigation, lesions of the anterior hippocampus interfere with stress responses and learning of emotional behavior. For example, in animals, activating the anterior hippocampus prevents the learning of stimulus-fear conditioning, such as the association of an auditory tone with an electrical shock. However, such lesions leave intact the process of context-fear conditioning: the animals can still make use of the posterior hippocampus to learn that a particular place (like the training cage) is associated with shocks (reviewed in Fanselow & Dong, 2010). In this respect, the anterior hippocampus functions more like the amygdala in learning and recalling the emotional value of a specific kind of outside world event, rather than a specific place. So we can almost imagine a continuum of function from posterior hippocampus to anterior hippocampus to amygdala (FIGURE 13.19): a

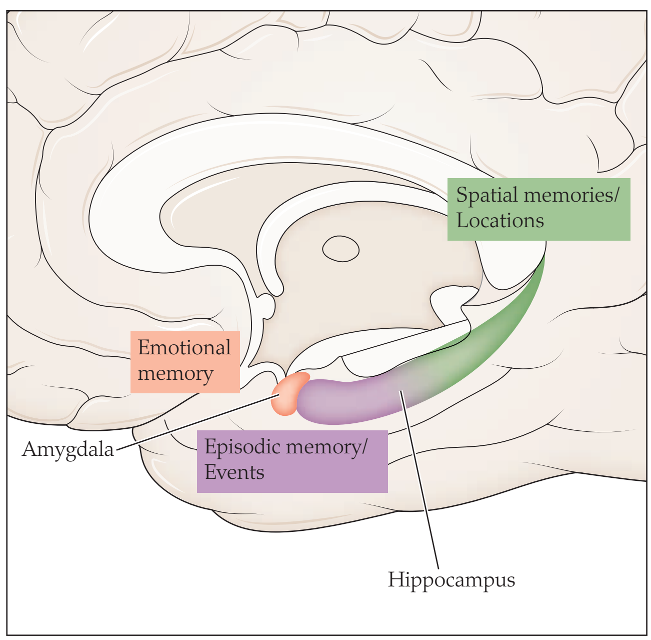

FIGURE 13.19 **The amygdala and hippocampus and their contributions to memory.** The amygdala and anterior hippocampus are important for emotional episodic memories, and the posterior hippocampus is important for spatial memories and navigation.